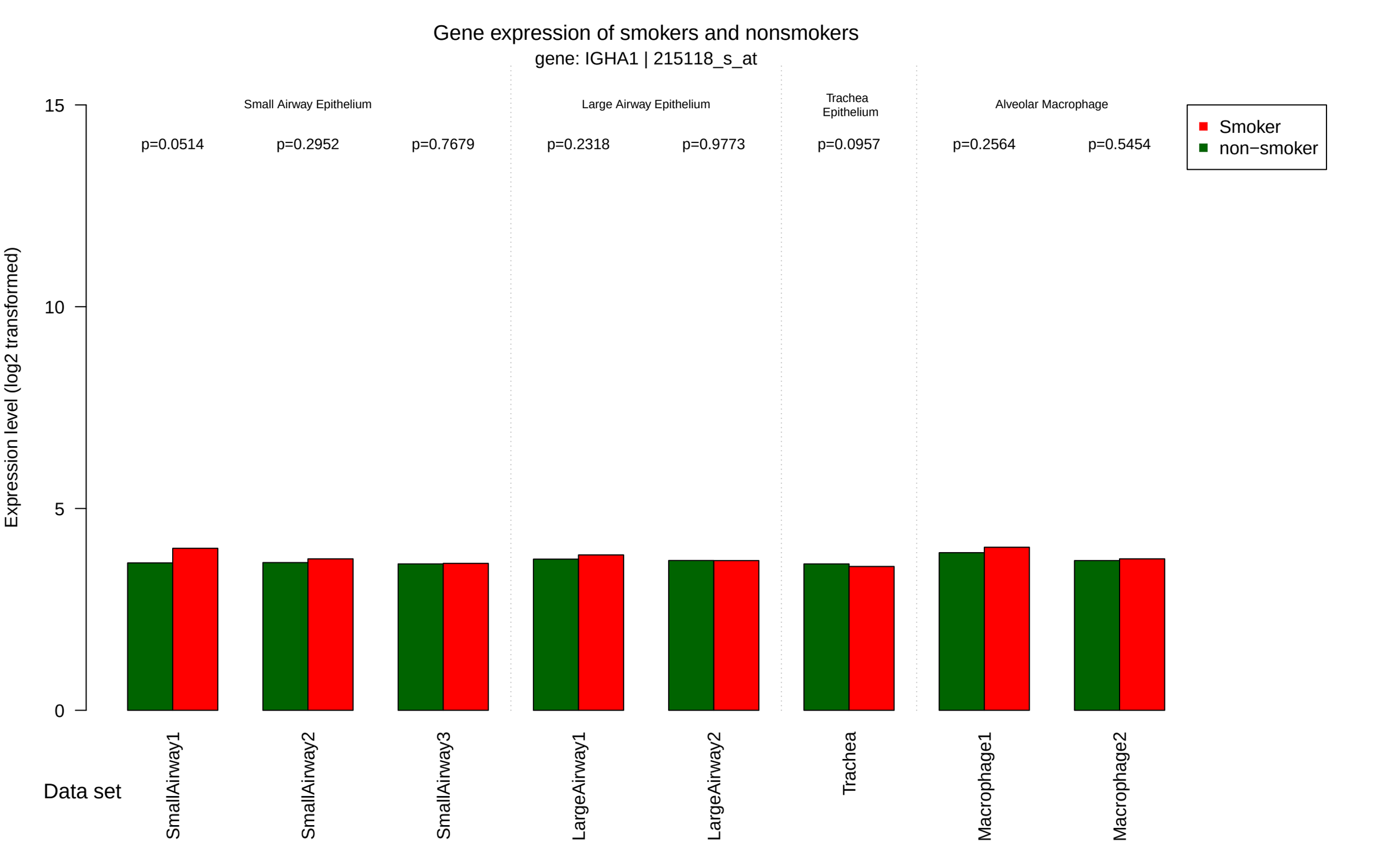
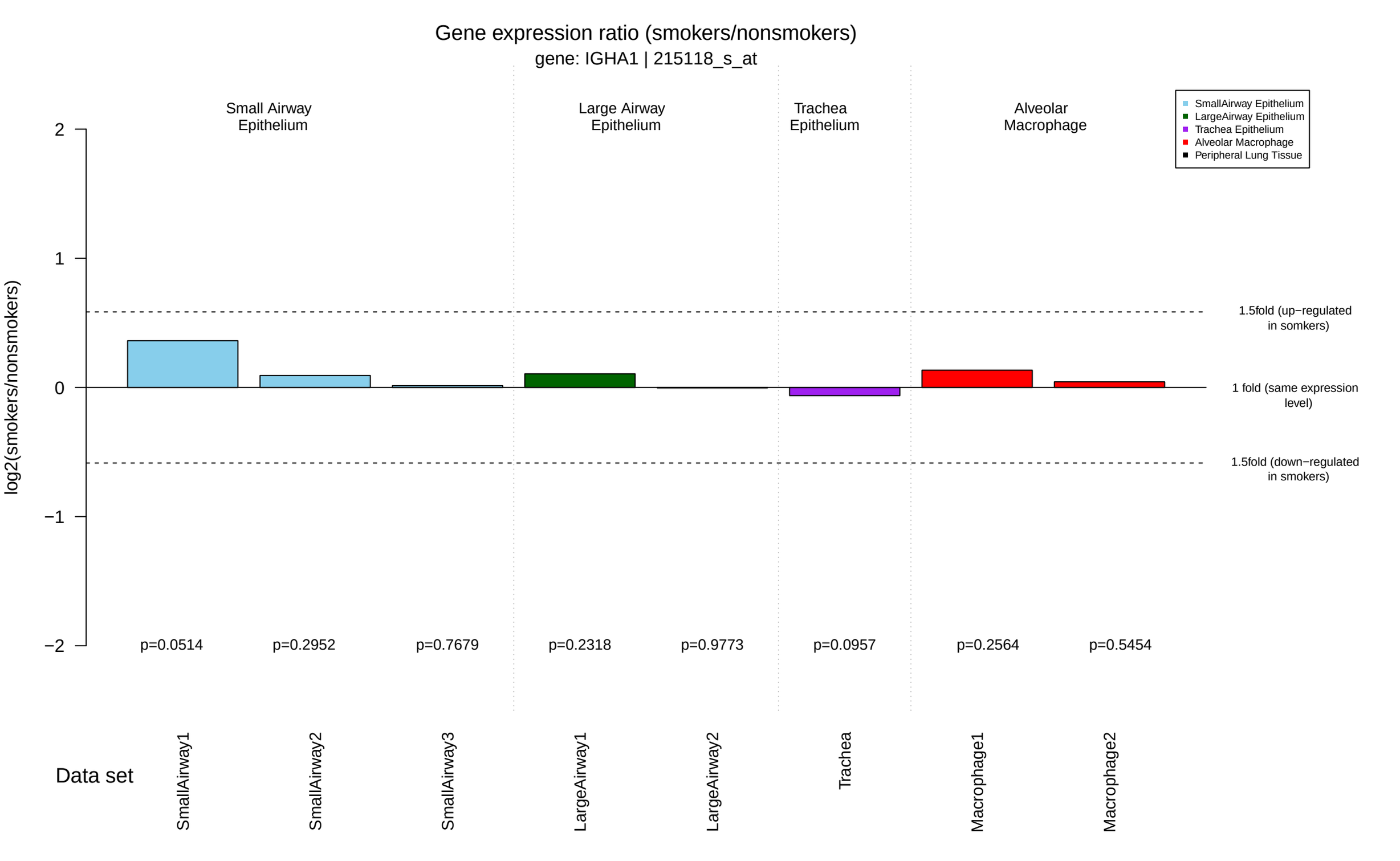
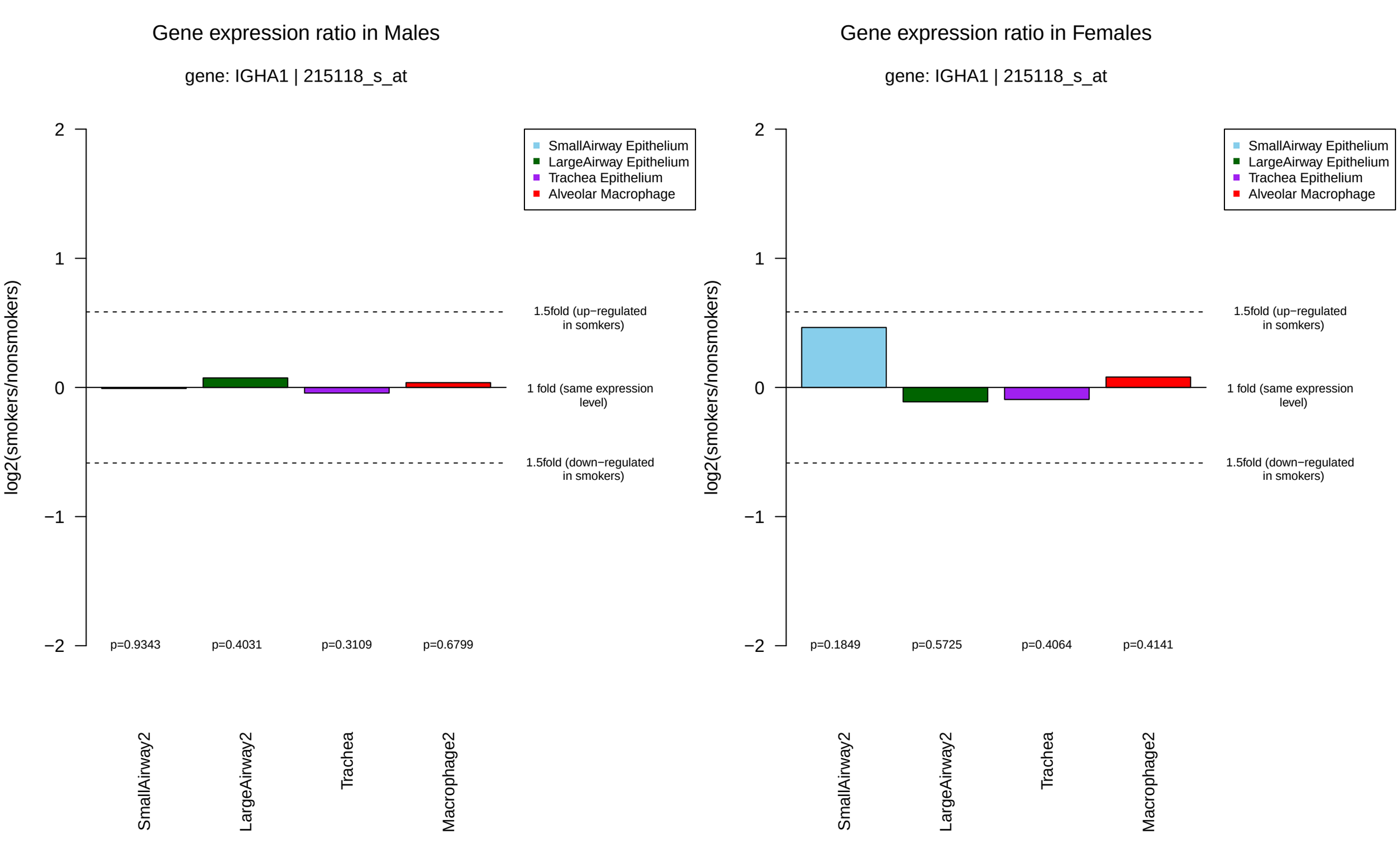
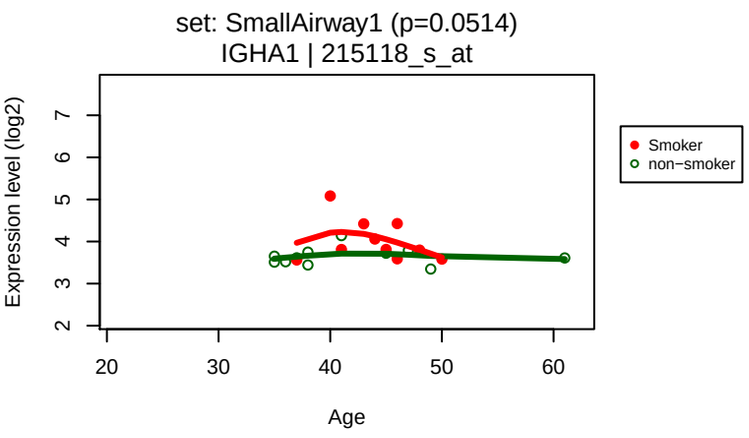
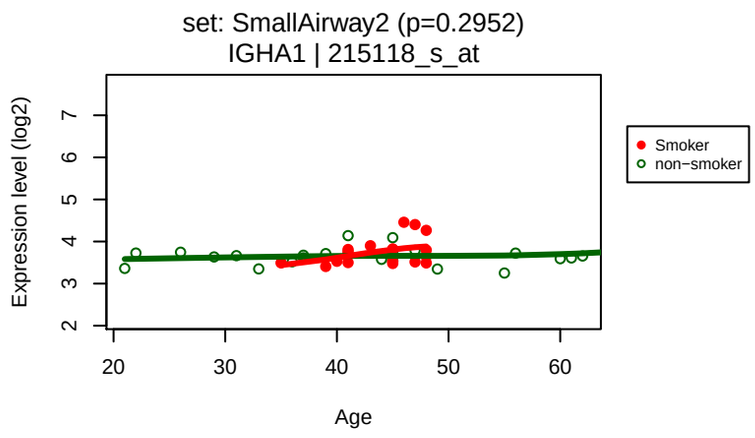
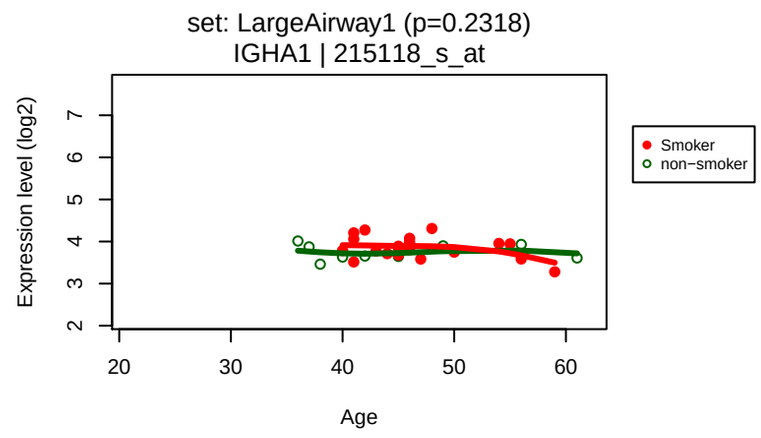

Gene expression of smokers and nonsmokers
gene: IGHA1 | 215118_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0514
p=0.2952
p=0.7679
p=0.2318
p=0.9773
p=0.0957
p=0.2564
p=0.5454
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

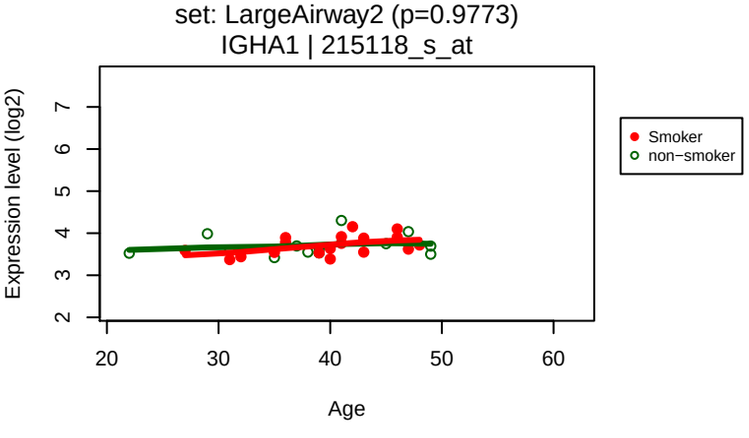

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 | 215118_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0514
p=0.2952
p=0.7679
p=0.2318
p=0.9773
p=0.0957
p=0.2564
p=0.5454
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

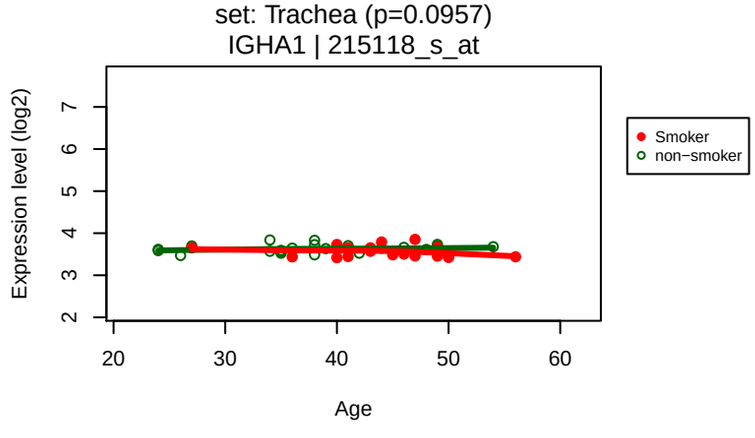

Gene expression ratio in Males
gene: IGHA1 | 215118_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9343
p=0.4031
p=0.3109
p=0.6799
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 | 215118_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1849
p=0.5725
p=0.4064
p=0.4141
SmallAirway2
LargeAirway2
Trachea
Macrophage2

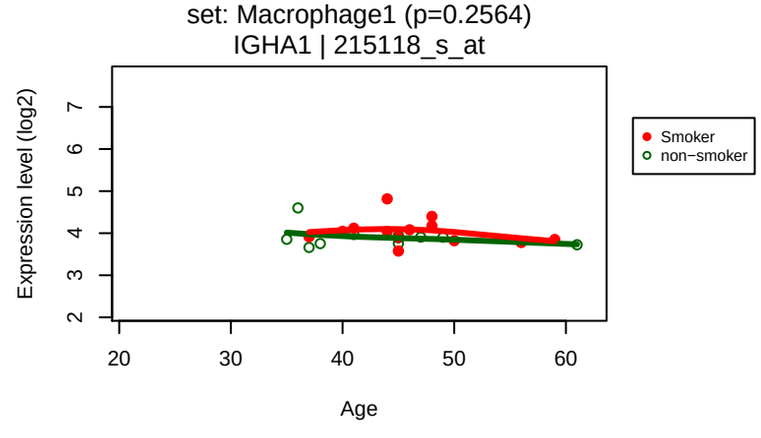

set: SmallAirway1 (p=0.0514)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker

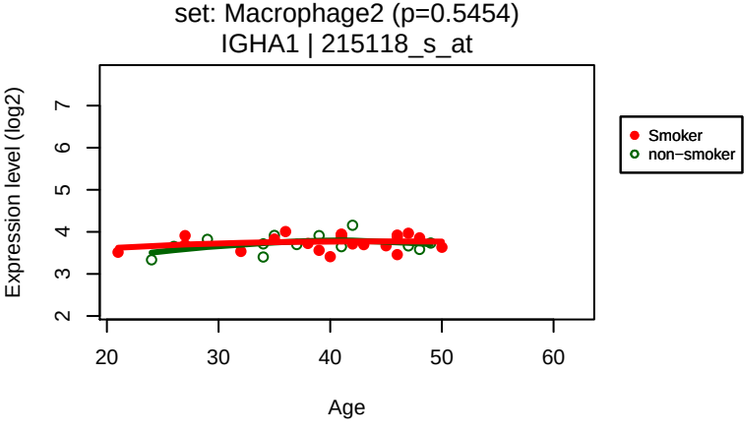

set: SmallAirway2 (p=0.2952)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker

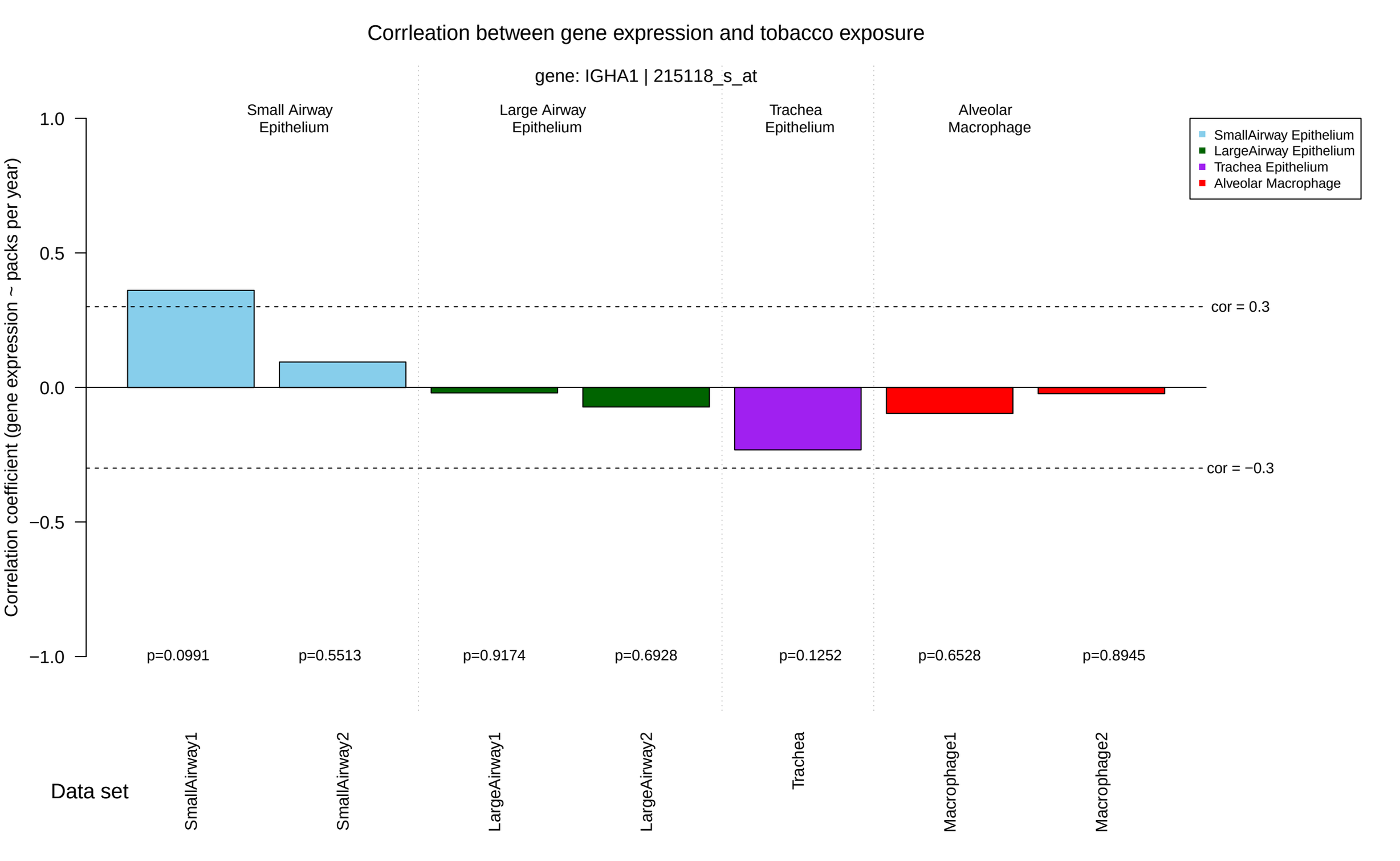

set: LargeAirway1 (p=0.2318)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9773)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0957)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2564)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5454)
IGHA1 | 215118_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 | 215118_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0991
p=0.5513
p=0.9174
p=0.6928
p=0.1252
p=0.6528
p=0.8945
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2